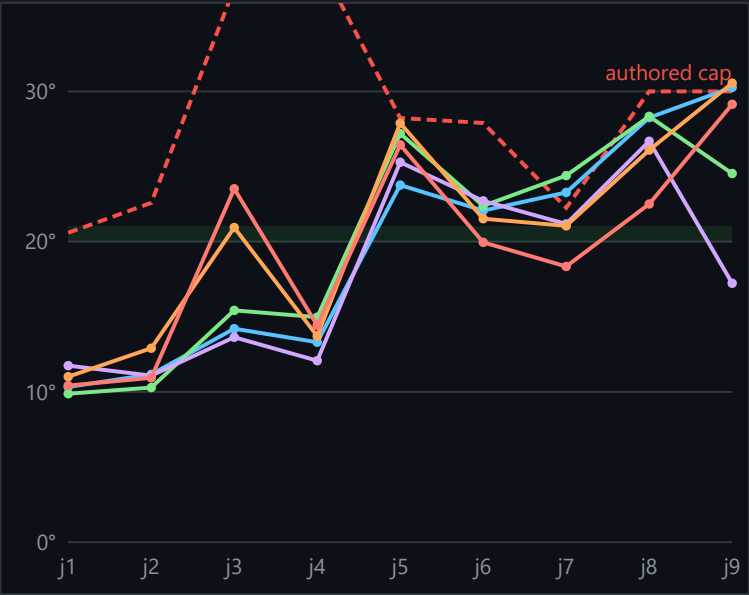


D-T2 · 5.1b — cutting drive is exhausted

One lever per run, everything else the approved MuJoCo base swim + head isolation, legs 0.1 kg. Gate: every joint under its own cap, bend spread down, girdle ratio toward 1.00.

Peak bend per spine joint (degrees)



Red dashed = the printed model's real range of motion (frozen, not a lever). Green band = the 20–21° uniform target. **No variant gets every joint under its cap.** Cutting the tail only lowers j8–j9 and leaves j7 (cap 22°) on the limit; cutting the hip pushes j6 up onto its cap instead and doubles the roll buzz. The front stays flat at 10–15° throughout.

Per variant

tail 0.60 — baseline (D-T2 best so far)

FAIL — clips j7,j8,j9

bend spread **19.9°** max/min **2.93**
girdle hind/front **0.94** speed **0.87 u/s**
roll 6.0/s at peak 0.80°

tail 0.40 — cut the tail harder

FAIL — clips j7,j8,j9

bend spread **18.5°** max/min **2.87**
girdle hind/front **1.07** speed **0.79 u/s**
roll 5.9/s at peak 0.93°

tail 0.25 — cut the tail hardest

FAIL — clips j7,j8

bend spread **15.6°** max/min **2.41**
girdle hind/front **0.96** speed **0.73 u/s**
roll 5.7/s at peak 0.94°

hip 0.75 — cut the hip instead

FAIL — clips j6,j7,j9

bend spread **19.5°** max/min **2.77**
girdle hind/front **1.10** speed **0.61 u/s**
roll 8.7/s at peak 1.55°

hip 0.55 — cut the hip hard

FAIL — clips j6

bend spread **18.7°** max/min **2.80**
girdle hind/front **1.13** speed **0.45 u/s**
roll 11.5/s at peak 1.59°

What this says. The front (j1–j4, 10–15°) never rises no matter how much tail is removed, so no downward-only profile reaches a uniform 20–21°. The next lever has to *lift the front* (nose/shoulder above 1.0), not cut further.